ASHOK KUMAR KUMARAVEL

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SUMMARY

Biotechnology and Microbiology Expert with a focus on molecular biology, metabolic engineering, and nanomaterials. Delivered sustainable technologies through research initiatives in wastewater treatment, environmental management, and biomaterial innovation.

EDUCATION

- **Ph.D. Chemical Engineering** (**Microbial Biotechnology**) **(94%), University of Ulsan, Ulsan, South Korea. (FEB 2024).**
- **Master of Science (M.Sc.) (Biotechnology) (74%), Bharathiar University, Tamil Nadu, India. (APR 2018).**
- **Bachelor of Science (B.Sc.) (Biotechnology) (74%), Bharathiar University, Tamil Nadu, India. (APR 2014).**

RESEARCH INTEREST AND EXPERTISE:

- **Molecular techniques**- Molecular cloning, Gene addition/Deletions/ mutations, Metabolic engineering through genetic engineering in microbes, microbial cell culture, primer designing, PCR, qPCR, AGE, SDS-PAGE and proficient in operating bioreactors.
- **Material Synthesis –** Green, Chemical and Microbial Synthesis of Nanoparticle.
- **Spectroscopic and analytical techniques** - ICP-OES (Inductively coupled plasma-optical emission spectrometry), XRD, FTIR, Raman spectroscopy, XPS, TEM, SEM, Zeta Analyzer, HPLC, GC-MS.
- **Software packages** – Python, R programming, Molecular Docking, Online Bioinformatic tools, Origin, Image and Diameter J, X'Pert high score plus, Microsoft Office.
- **GitHub Projects –** Codon Optimization& Bioinformatics (Finished), Metabolic Engineering Toolkit (Dev), Autonomous Molecular Docking tool (Dev), Autonomous Peptide-Metal Ion Binding prediction with DFT tool(Dev).

PHD RESEARCH SUMMARY (MAR 2019 – FEB 2024)

During my PhD at the University of Ulsan under Prof. Soon-Ho Hong, I developed a microbial cell surface display (MCSD) system by engineering *Escherichia coli* to remove cobalt from water using surface-expressed binding peptides. The work also enabled the synthesis of cobalt oxide nanoparticles from bio adsorbed cobalt, which showed effective dye degradation, drug degradation and promising anticancer potential. This research combines molecular biology, environmental engineering, and nanotechnology to deliver sustainable solutions for environmental remediation and biomedical applications.

RESEARCH FELLOW (NTU) (NOV 2024 – APR 2025)

- At National Taiwan University, I worked on transforming agricultural waste (peanut shell powder) into enhanced bacterial cellulose (BC) for packaging applications.
- Through chemical and enzymatic modifications, the study achieved ~40% improvement in tensile strength and mechanical properties of BC, demonstrating its commercial viability.

Post-Doctoral Researcher (UNIST) (Aug 2025 - Feb 2026)

As a Postdoc at Ulsan National Institute of Science and technology, I did a national project focused on engineering *Pseudomonas denitrificans* for efficient biosynthesis of Poly(3-hydroxypropionate) from glycerol and gluconate, contributing to sustainable biopolymer production.

TEACHING EXPERIENCE

- Two years as a Teaching Assistant during PhD (Biotechnology, Biochemistry, Molecular Biology, Genetics).
- Delivered ~15 hours of lectures on biomaterials, modifications, and applications during my research fellowship at NTU.

PRESENT:

- **Adjunct Professor, (Since Dec,19 - 2025)** at Centre for Research and Innovation, Academy of Maritime Education and Training (AMET) Deemed to be University, Kanathur, Chennai 603112, Tamil Nadu, India.

AWARDS AND GRANTS

- **Brain Korea 21 (BK21)** - National Scholarship for Outstanding Scholars (Apr 2020 – Mar 2022).
- **Low Carbon Green Energy Project** – Ulsan, Korea (Jul 2022 – Feb 2024).
- **National Science and Technology Council (NSTC)** -Taiwan Postdoctoral Scholarship (NOV 2024 – MAY 2025): "Optimizing the production of components that inhibit melanin production in *Ganoderma lucidum* mycelium using PCS bioreactor".

PATENT

Ashok Kumar Kumaravel, Hong-Soon Ho, Kang-Sung Gu. "Synthesis of cobalt oxide nanoparticles by recombinant microorganisms with modified cell surface and their use" **South Korea Taebaek Patent Firm: 9-2008- 100101-3. Application No: 1-1-2023-1017641-54 / 10-2023-0122295 / 2023.09.14.**

PUBLICATIONS (SCI / ESCI Indexed) (12 Publications)

2026

- **Ashokkumar Kumaravel**[1], Saranya Shanmugasundarama, Kim-Ngan T. Trana, Palanivel Velmurugan, Soon Ho Hong[*], "Cobalt oxide nanoparticle synthesis by cell surface engineered recombinant Escherichia coli and the potential application on photocatalytic degradation of norfloxacin", **Results in Engineering, (IF-7.9) (Q1)**
- Saranya Shanmugasundaram, **Ashokkumar Kumaravel**[2], Soon Ho Hong[*]. "*E. coli* Surface-Displayed Nickel-Binding Peptides for Preferential Nickel Recovery from Battery Wastewater: Computational Modelling and Experimental Validation", **Journal of Environmental Management, (IF-8.4) (Q1)**

2025

- **Ashokkumar Kumaravel**[1], Shin-Ping Lin[1], Shella Permatasari Santoso, Saranya Shanmugasundaram, Hsien-Yi Hsu, Chang-Wei Hsieh, Yu-Chieh Chou, Hui-Wen Lin, Kuan-Chen Cheng*, Unlocking the potential of bacterial cellulose: synthesis, functionalization, and industrial impact, **International Journal of Biological Macromolecules, (IF-8.5) (Q1)**
- Bharat Bhargawa[1], Tran Van Tam[1], **Ashok Kumar Kumaravel**[2], Sandeep Kumar Lakhera, Soon Ho Hong, Won Mook Choi, Ik-Keun Yoo*, "Optimized g-C_3N_4/TiO_2 composites for visible-light-driven Sono photocatalysis in norfloxacin removal" – **Environmental Research. (IF-7.7) (Q1)**

2024

- **Ashokkumar Kumaravel**[1], Selvamani, V., Sengupta, T., Gu Kang, S., & Ho Hong, S*. (2024). Surface- engineered recombinant *Escherichia coli* for the potential application of the cobalt-contaminated wastewater treatment and the photocatalytic dye degradation. **Bioresource Technology. (IF-9) (Q1)**
- **Ashokkumar Kumaravel**[1]**,** Sengupta, T., Sathiyamoorthy Padmanaban, J Jeong, Gu Kang, S., Ho Hong, S*. "Cobalt oxide nanoparticle synthesis by cell-surface engineered recombinant *Escherichia coli* and potential application for anticancer treatment". **ACS omega (IF-4.3) (Q1)**

2023

- **Ashok Kumar Kumaravel**[1], Vidhya Selvamani[1], SH Hong*, "Photocatalytic reduction of methylene blue by the cobalt-bound surface-engineered recombinant *Escherichia coli* as whole-cell biocatalyst," **Bioengineering, (IF-3.7) (Q2)**
- **Ashok Kumar Kumaravel**[1], KNT Tran[1], SH Hong*, "Impact of the Synthetic Scaffold Strategy on the Metabolic Pathway Engineering," **Biotechnology and Bioprocess Engineering, (IF-3) (Q2)**

2022

- KNT Tran[1], **Ashok Kumar Kumaravel**[2], J Jeong[3], SH Hong*, "High Yield Fermentation of L-serine in Recombinant *Escherichia coli* via Colocalization of SerB and EamA through Protein Scaffold," **Biotechnology and Bioprocess Engineering, (IF-3) (Q2)**
- Nivedhitha Kabeerdass[1], Karthikeyan Murugesan[2], Natarajan Arumugam[3], Abdulrahman I Almansour[3], Raju Suresh Kumar[3], Sinouvassane Djearamane[4], **Ashok Kumar Kumaravel**[5] et.al, "Biomedical and Textile Applications of *Alternanthera sessilis* Leaf Extract Mediated Synthesis of Colloidal Silver Nanoparticle", **Nanomaterials, (IF-4.4) (Q1)**
- I.Jenish[1], A Felix Sahayaraj[2], V Suresh[3], J Mani Raj[2], M Appadurai[4], E Fantin Irudaya Raj[5], Omaima Nasif[6], Saleh Alfarraj[7], **Ashok Kumar Kumaravel**[8], "Analysis of the hybrid of mudar/snake grass fiber- reinforced epoxy with nano-silica filler composite for structural application," **Advances in Materials Science and Engineering, (IF-2.272) (Q2)**

2021

- S Somasundaram[1], J Jeong[2], **Ashok Kumar Kumaravel**[3], SH Hong*, "Whole-cell display of *Pyrococcus horikoshii* glutamate decarboxylase in *Escherichia coli* for high-titer extracellular gamma-aminobutyric acid production," **Journal of Industrial Microbiology and Biotechnology, (IF-3.2) (Q1)**

UNDER REVIEW

- Chandran, Likha[1]; **Kumaravel, Ashokkumar[1];** Yohan, Ravi Kumar; Tota, Jagadish; Jagannathan, Mohanraj; Ben Moussa, Sana; Alzahrani, Abdullah, "Electro spun Polymeric Biomaterials: From Fundamental Design to Advanced Biomedical Applications". – **Advanced Industrial and Engineering Polymer Research (IF-12) (Q1)**.
- **Ashokkumar Kumaravel[1],** Sathiyamoorthy Padmanaban, Saranya Shanmugasundaram Saranya, Likha Chandran; "Valorization of Peanut Shell Waste into High-Performance Bacterial Cellulose: Comparative Evaluation of Pretreatment Strategies and Process Integration" – **International Journal of Biological Macromolecules (IF-8.5) (Q1).**
- **Ashokkumar Kumaravel[1],** Sathiyamoorthy Padmanaban, Saranya Shanmugasundaram Saranya, Bharat Bhargawa; "Phytochemical profiling and multifunctional biodiversity of Acorus calamus: Isolation of Isocaespitol and its antidiabetic and anti-inflammatory potential via Invitro and Insilco approaches" **Frontiers in Pharmacology (IF-4.8) (Q1)**
- **Ashokkumar Kumaravel[1],** Sathiyamoorthy Padmanaban, Saranya Shanmugasundaram Saranya, Senthamil Selvi Poongavanam; "Development and Evaluation of a Yogurt-Based Functional Formulation Enriched with *Glycyrrhiza glabra* and *Solanum nigrum*". **Journal of Functional Foods (IF-4) (Q1)**

BOOK CHAPTER – PUBLISHED

- Sriya.D, **Ashokkumar Kumaravel**, Saranya Shanmugasundaram. "Microbes in Water, Sanitation, and Public Health: A Data-Driven Analysis of Pathogen Dynamics and Intervention Efficacy"- **Scopus Indexed**

ON PROGRESS

- **Ashokkumar Kumaravel**, Saranya Shanmugasundaram, Likha Chandran, and Senthamil Selvi Poongavanam. "The Role of Fusobacterium in Biofilm Development: A Central Bridge to Pathogenesis" - **Scopus Indexed**
- **Ashokkumar Kumaravel**, Likha Chandran, Saranya Shanmugasundaram. "Hidden Threats: Chemical and Biological Contaminants in water and their adverse impact in human health" - **Scopus Indexed**
- Sriya.D, **Ashokkumar Kumaravel**, Saranya Shanmugasundaram. "Microplastics and Organic Pollutants: Accumulation, Bioavailability, and Ecological Consequences in Coastal and Estuarine Systems"- **Scopus Indexed.**

- Kotthapalli Prashanth[1], **Ashokumar Kumarvel**[2,3], Y. Aparna[4], S. Anju[4], Laurent Dufosse[5], Jamil Talukder[6,] "Sumana Kumar[1]MicroRNA-Edited Gut Microbiota as an Epigenetic Vector" - **Scopus Indexed.**
- **Ashokkumar Kumaravel**, K. Anuradha, Shalini Rachel, Kotthapalli Prashanth, Sumana Kumar. Post-COVID-19 Era: A Bloom in Antibiotic Resistance in Microbial Community - **Scopus Indexed.**

CONFERENCE PROCEEDINGS

2025

- Vivek Kumar Gaur, Tayyab Islam, Thuan Phu Nguyen-Vo, Miri Kim, Satish Kumar Ainala, **Ashok Kumar Kumaravel**, Payal Mukherjee, Sakshi, Sunghoon Park, "Metabolic engineering of Escherichia coli for efficient production of poly (3- hydroxypropionate-co-3-hydroxybutyrate) copolymer with tailored monomer composition" **IJ-Bio 5-8 /11/2025**.

2023

- **Ashokkumar Kumaravel,** Jh Jeong, SH Hong, "*Colocalization of SerB and EamA through Protein Scaffold for the High-Yield Fermentation of L-serine in Recombinant Escherichia coli*," **Korean Society of Biological Engineering conference, 2023/4, 745.**
- Jh Jeong, **Ashokkumar Kumaravel,** SH Hong, "Displaying *Neurospora crassa* Glutamate Decarboxylase on the Surface of *Escherichia coli* for the Extracellular Gamma-aminobutyric Acid Production from High Cell Density Culture," **Korean Society of Biological Engineering conference, 2023/4, 746.**
- **Ashokkumar Kumaravel,** Jh Jeong, SH Hong, "High Yield Fermentation of L-serine in Recombinant *Escherichia coli* by Colocalization of Pathway Enzymes through the Protein Scaffold," **Korean Society of Biological Engineering conference. 2023/10, 886.**
- Jh Jeong, **Ashokkumar Kumaravel**, SH Hong, "Modification of the *Escherichia coli* Cellular Surface for the Extracellular Gamma-aminobutyric Acid Production from High Cell Density Culture", **Korean Society of Biological Engineering conference. 2023/10, 887.**

2022

- **Ashokkumar Kumaravel,** Jh Jeong, SH Hong, "Cell Surface Display of *Pyrococcus horikoshii* Glutamate Decarboxylase in *Escherichia coli* for High-titer Extracellular Gamma-aminobutyric Acid Production," **Korean Society of Biological Engineering conference, 2022/4, 546.**
- Jh Jeong, **Ashokkumar Kumaravel,** SH Hong, "Construction of Pesticides Recovering *Escherichia coli* through the Cell Surface Display of Binding Peptide," **Korean Society of Biological Engineering conference, 2022/4, 365.**

2021

- **Ashokkumar Kumaravel,** Jh Jeong, SH Hong, "Enhanced Production of L-serine in *Escherichia coli* Through Synthetic Protein Scaffold of SerB, SerC, and EamA", **Korean Society of Biological Engineering conference, 2021/4, 583.**
- Jh Jeong, **Ashokkumar Kumaravel,** Kim-Ngan, Sh Hong. "Improved Itaconic Acid Production by Employing Protein Scaffold Between GltA, AcnA, and CadA in Recombinant *Escherichia coli*," **Korean Society of Biological Engineering conference, 2021/4, 596.**
- **Ashokkumar Kumaravel,** Jh Jeong, SH Hong, "Novel Strategy for Malic Acid Production in Recombinant *Escherichia coli* via Protein Colocalization," **Korean Society of Biological Engineering conference, 2021/10, 431.**
- Jh Jeong, **Ashokkumar Kumaravel,** SH Hong, "Development of Fenitrothion Removing Recombinant *Escherichia coli* by the Cell Surface Display of Pesticide-binding Peptide," **Korean Society of Biological Engineering conference, 2021/10, 432.**